

B.2 Lab 2: Functions, Vectorization, and Quantifying Errors

Problem 1: Approximating π

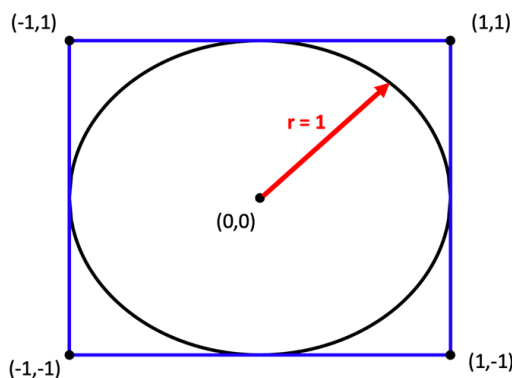
Last week we used iteration to visualize the unit circle and grow our own ferns. In this exercise, we will perform an iteration through *vectorization* to approximate π . We know the area of a circle is:

$$\text{Area} = \pi * r^2$$

where r is the radius. We also know the algebraic formula for a circle centered on the origin:

$$x^2 + y^2 = r^2$$

Choosing $r = 1$ gives us the formula for the unit circle. Let's draw the unit circle and also the square, S , that circumscribes it:



Recall that MATLAB's `rand` function returns numbers drawn from the uniform distribution on the interval $[0, 1]$. If we choose a point within S completely at random, what is the probability that it will land inside the circle? What must we do to `rand`'s output to ensure we draw random numbers from within S ? Once you're confident that you've answered these questions, take a look at `piApprox.m`.

`piApprox` implements an approach to area estimation known as the “hit-or-miss” method, which is itself a type of Monte Carlo Integration Method.³ Monte Carlo methods can be used to solve challenging problems in biomedical engineering, such as calculating the dose delivered from cancer radiotherapy.⁴

Problem 1.1: Write a function called `piApproxVectorized.m` that achieves the same goal as `piApprox.m` but without the use of a for-loop.

Problem 1.2: Write a function named `piApproxError.m` that uses a for-loop to call your `piApproxVectorized` for $n = 10, 100, 1000, \dots, 10^6$. Within the loop, compute and store the *relative error* for each n . To do this, we need a “true solution” for π . Use MATLAB's built-in value for `pi` as the true value (type `pi` into the command line to see this value). Your function should create a plot of the error as a function of n using `loglog`. What's the relationship between the accuracy of our approximation and the total number of points used to generate it? Provide an answer to this question in the header's program description.

Note: Please include the following line of code before you call the `rand` function: `rng(0)`; This will ensure that we all get the same answers.

³https://en.wikipedia.org/wiki/Monte_Carlo_integration

⁴<https://ro-journal.biomedcentral.com/articles/10.1186/s13014-018-1065-3>

Problem 2: ???

The objective of this problem is to calculate the area of an unknown object. Fortunately, we have a function `oracle.p` that takes as input 2D points within the square $[3, 14] \times [3, 14]$ and returns a '1' if the point is within the object and a '0' if it is not. Download `oracle.p` and enter the command » `help oracleHelp`. Membership oracle functions are a tool used to calculate the volume of high dimensional shapes. You can learn more about them in the aptly named paper “Computing the Volume is Difficult.”⁵

Devise an algorithm for calculating the area of the shape defined by `oracle.p` and write a function `areaCalc.m` that implements your approach. Your `areaCalc` should generate a plot of the object with the calculated area displayed in the title. Be sure to include a description of your algorithm in your function header.

File Naming

For ease of grading, use the following naming scheme for your files:

<code>[FirstName][LastName]_[filename].m</code>	For programs
<code>[FirstName][LastName]_[filename]_plot_1.pdf</code>	For corresponding plots (export them to pdf)
<code>[FirstName][LastName]_[filename]_plot_2.pdf</code>	For problem with more than one plot

Submission Checklist

- Problem 1.1: `piApproxVectorized.m`
 - Fully commented code
 - Correct use of vectorized random numbers
 - Correct use of vectorized if-statement
- Problem 1.2: `piApproxError.m`
 - Fully commented code
 - Accurate description of error with respect to n
 - Correct use of for-loop
 - Correct output plot and title
- Problem 2: `areaCalc.m`
 - Header description of your algorithm
 - Fully commented code
 - Correct plot of the unknown object
 - Correct code and area calculation

⁵<https://link.springer.com/content/pdf/10.1007/BF02187886.pdf>